

Experimental Design Report: Candy Color Sorter

Name/Candidate number: Liu Moxin , 299102

DATE : 2025.5.8

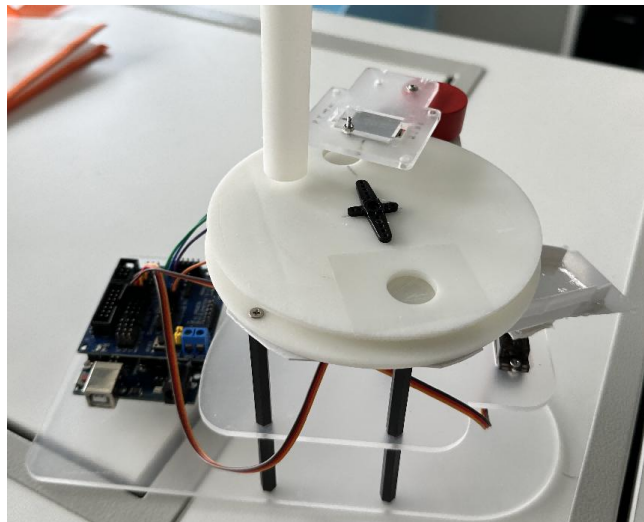
1. Introduction

Project Background

This design aims to build an automated candy bean sorting system based on color recognition. The system utilizes the TCS34725 sensor for color detection and the MG90S servo to control the sorting path. It is designed to meet the practical requirements of low cost, high efficiency, and easy scalability to adapt to different application scenarios.

Design Objectives

- **Color Recognition Accuracy:** Achieve or exceed 95% to ensure accurate sorting.
- **Sorting Speed:** No less than 1 candy bean per 3 seconds to improve production efficiency.
- **Cost Control:** Keep the total cost strictly within 300 yuan (the experimental budget was 1000 yuan, and the actual expenditure was 293.2 yuan) to achieve high cost-effectiveness.



1-1 picture of real products

2. Overall Design

Communication Method

This system adopts the I2C communication method to realize data transmission between components. The Arduino UNO R3, as the main control board, is connected to the TCS34725 color sensor via the I2C bus to obtain color detection data. Meanwhile, the Arduino UNO R3 uses PWM signals to control the MG90S servo, achieving precise control over the candy bean sorting path. I2C communication is simple and efficient, capable of meeting the system's requirements for data transmission speed and stability.

Key Technical Implementations

1. Image Processing Technology (Color Recognition)

- **Implementation Method:** Utilize the TCS34725 color sensor to read the RGB values of candy beans. The sensor features high precision (16-bit ADC) and adjustable gain (4X), with a built-in IR filter, effectively adapting to complex lighting environments. When reading RGB values, an integration time of 50ms is set to ensure accurate color detection.
- **Training Process:** Preliminary training was conducted in a dormitory environment. Different-colored candy beans were placed in front of the sensor, and the corresponding RGB value ranges for each color were recorded. Since the lighting conditions in the dormitory were relatively stable, preliminary color intervals were obtained. However, considering that lighting conditions might change in actual usage scenarios, a light-blocking device was designed to prevent sensor recognition errors. Before taking the system to the classroom for actual testing, five tests were conducted. In each test, 10 candy beans of each color were selected. Based on the test results, the color intervals were adjusted, and accurate color thresholds were finally determined. For example, for brown, the threshold was set as R:400 - 600, G:400 - 620, B:300 - 500, etc.

2. Sorting Technology

- **Implementation Method:** This system adopts a dual-servo collaborative design scheme. The top servo (pin 7) is responsible for rotating the candy bean tray to the detection position ($175^{\circ} \rightarrow 100^{\circ}$), simulating the process of candy beans falling. The bottom servo (pin 8) precisely controls the slope angle ($10^{\circ} \sim 160^{\circ}$) according to the color detection results, guiding the candy beans to fall accurately into the designated position. During the sorting process, reasonable timing control is employed to ensure that the top servo pauses for color detection, the bottom servo rotates the slope based on the preset color threshold judgment results, the candy beans slide naturally into the designated container under

gravity, and finally, the top servo quickly resets ($5^{\circ}\rightarrow175^{\circ}$) to prepare for the next sorting.

Two Important Issues in the Design

- 1. **Image Processing Technology:** Accurately recognizing candy beans of different colors is one of the key issues in this design. Since changes in lighting conditions can affect color detection, it is necessary to design a light-blocking device and adjust the color intervals through a large number of actual tests to improve the accuracy of color recognition.
- 2. **Sorting Technology:** Ensuring that candy beans can be accurately and smoothly sorted into the designated positions is another important issue. In terms of mechanical structure design, it is necessary to consider the selection of servos and their collaborative working methods, as well as solve the problem of candy beans getting stuck on the slope to ensure the smoothness of the sorting process.

3. Hardware Design

3.1 Hardware Selection and Cost Analysis

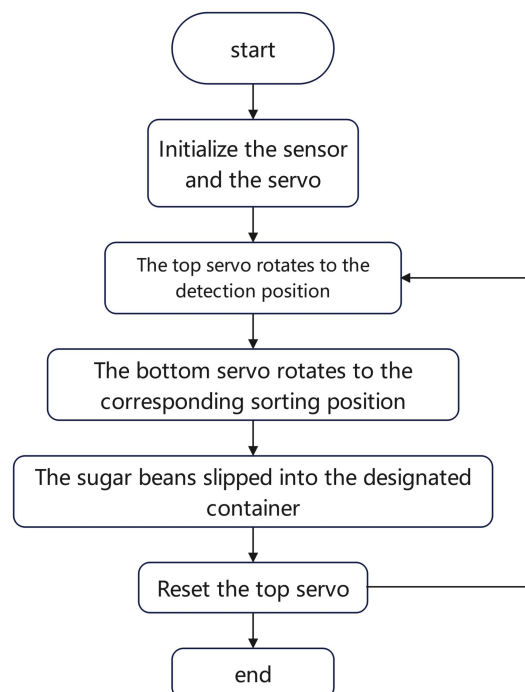
Material	Unit Price (Yuan)	Quantity	Total Price (Yuan)	Selection Reason
MG90S Servo	10.33	2	20.66	High cost-effectiveness, light weight (9g), fast response speed (0.1s/60°), very suitable for light-load sorting tasks.
TCS34725 Color Sensor	11.01	2	22.02	Features high precision (16-bit ADC), adjustable gain (4X), and a built-in IR filter, effectively adapting to complex lighting environments.
Arduino UNO R3 Kit	146.6	2	293.2	Strong compatibility, rich development resources, supports PWM/I2C communication, facilitating system integration and development.
ABS Plastic Plate	11.21	1	11.21	High mechanical strength, easy to process, suitable for making sorting channels and brackets to meet the system's structural requirements.
Total	-	-	293.2	-

3.2 Mechanical Structure Design

Sorting Logic

This system adopts a dual-servo collaborative design scheme:

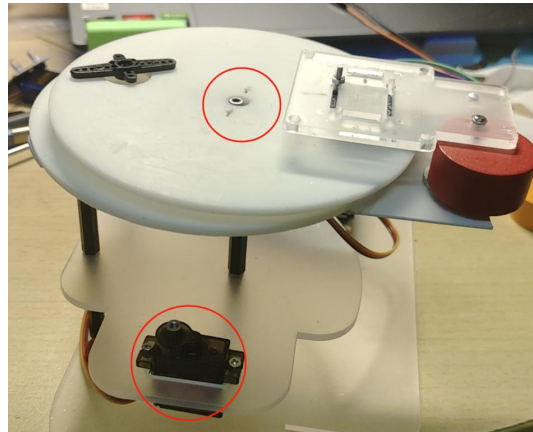
- **Top Servo (pin 7):** Responsible for rotating the candy bean tray to the detection position ($175^{\circ} \rightarrow 100^{\circ}$), simulating the process of candy beans falling.
- **Bottom Servo (pin 8):** According to the color detection results, precisely controls the slope angle ($10^{\circ} \sim 160^{\circ}$) to guide the candy beans to fall accurately into the designated position.



3-1 Sorting flowchart

Why Choose 180° Servos?

- **Angle Coverage:** The 180° servo (MG90S) can fully cover the angle range required for sorting ($5^{\circ} \sim 175^{\circ}$), and its cost is only 1/3 of that of a 360° servo, effectively controlling costs while meeting functional requirements.
- **Control Complexity:** A 360° servo requires an additional encoder to achieve precise position control, which not only increases the complexity of the system but also violates the principle of low cost.



3-2Steering gear indicator diagram

3.3 Analysis and Improvement of Candy Bean Sticking on the Slope

During the testing process, we found that the slope fixed with 3D printing and glass glue experienced candy beans getting stuck after multiple runs. After analysis, we speculated that it was because the sugar coating on the candy beans melted after being picked up by hand multiple times, increasing the friction between the candy beans and the slope, leading to the sticking phenomenon.

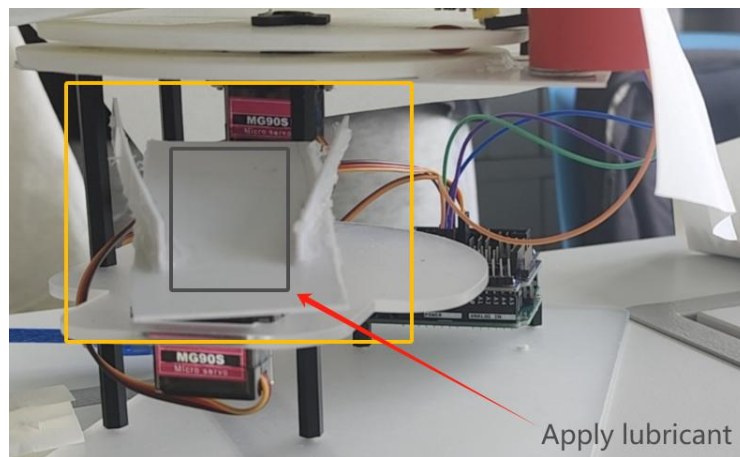
In response to this problem, we proposed two possible solutions: increasing the slope and increasing the slope lubricity.

- **Solution 1: Increase the Slope**
 - **Advantages:** Increasing the slope can rely on gravity to make the candy beans slide more smoothly, reducing the problem of sticking caused by friction. This method is simple and direct, requiring no additional materials or complex operations.
 - **Disadvantages:** If the slope is increased too much, it may cause the candy beans to slide too fast, exceeding the response speed of the bottom servo and affecting the sorting accuracy. At the same time, a larger slope will also increase the overall height of the system, and other components may need to be adjusted to adapt to the new structure.
- **Solution 2: Increase the Slope Lubricity**
 - **Advantages:** By applying a lubricant to the slope surface, the friction coefficient between the candy beans and the slope can be reduced, making it easier for the candy beans to slide. This method can solve the problem without changing the existing mechanical structure, having a small impact on the overall system.
 - **Disadvantages:** It is necessary to select an appropriate lubricant to ensure that it does not contaminate the candy beans or affect the sensor's detection results. In

addition, the lubricant may evaporate or be lost over time and needs to be supplemented regularly.

Final Solution Selection

After comprehensively considering the advantages and disadvantages of the two solutions, we decided to adopt the method of increasing the slope lubricity. First, we selected a food-grade lubricant that would not contaminate the candy beans or affect the sensor's detection. When applying the lubricant, we used a uniform application method to ensure that the entire slope surface was well lubricated. After actual testing, the problem of candy beans getting stuck on the slope was effectively solved, and the system's sorting process became smoother.



3-3 operation chart

3.4 Collaborative Principle of Sensor and Actuator

- **Color Detection Stage:** The top servo pauses operation, and the TCS34725 sensor reads the RGB values. The integration time is set to 50ms to ensure the accuracy of color detection.
- **Sorting Stage:** The bottom servo rotates the slope according to the preset color threshold judgment results (e.g., brown corresponds to 10°), and the candy beans slide naturally into the designated container under gravity.
- **Reset Stage:** The top servo quickly resets ($5^\circ \rightarrow 175^\circ$) to prepare for the next sorting.

4. Software Design

4.1 Key Code Analysis

```
// Color detection function (threshold method)

int readColor() {
    uint16_t r, g, b;
    tcs.getRawData(&r, &g, &b, &c);
    // Brown threshold (calibrated through experiments)
    if (r > 400 && r < 600 && g > 400 && g < 620 && b > 300 && b < 500) return 1;
    // ...Other color logic
}
```

- **Threshold Optimization:** Through a large number of experimental calibrations of the RGB range (e.g., brown R:400 - 600), the interference of ambient light on color detection is effectively reduced, improving the accuracy of color recognition.

4.2 Test Data and Problems

Initial Test

In the initial test stage, the system operated at a relatively slow speed (servo delay of 5000ms). A total of 50 candy beans were sorted, and the accuracy reached 100%, fully proving the stability of the system at low speeds.

Accelerated Test

In the accelerated test, the system sorted 76 candy beans in 3 minutes, but 2 errors occurred.

- **Error Cause:** The bottom servo did not respond. Further analysis revealed that the code logic did not wait for the servo to complete the action, leading to errors in the sorting process.

- **Improvement Measure (Code Modification):**

case 1:

```
bottomServo.write(10);
```

```
delay(200); // Increase the servo action waiting time to ensure that the servo completes the action before
proceeding to the next step
```

```
break;
```

5. Performance Analysis and Improvement

5.1 Current Scheme Bottlenecks

Speed Limitation:

- The single sorting cycle is approximately 2.5 seconds (1 second for detection + 1.5 seconds for reset). The theoretical maximum speed is only 24 candy beans per minute, which is far from meeting the production needs for higher efficiency.
- Hardware Performance Analysis: The MG90S servo motor used in the system has relatively low performance parameters. Its speed is limited, with a typical response time of 0.1s per 60° rotation. In the sorting process, the top servo needs to rotate between 5° and 175°, and the bottom servo needs to adjust between 10° and 160°. These rotations require multiple steps, and the limited speed of the MG90S servo leads to a significant portion of the time being consumed in the servo movement, thus restricting the overall sorting speed.
- Additionally, the torque of the MG90S servo is only 1.6kg·cm. When operating at higher speeds, the insufficient torque makes it difficult for the servo to overcome the inertia and friction of the mechanical structure, resulting in stuttering during the rotation process. This not only affects the smoothness of sorting but also further slows down the sorting speed.

Servo Response Delay:

- The MG90S servo has an inherent response delay. Due to its simple internal structure and control mechanism, it cannot respond immediately to the control signals from the Arduino UNO R3. In the code, if there is no proper waiting time after sending the servo control command, the subsequent operations may be executed before the servo completes its action, leading to errors in the sorting process.
- Hardware Performance Analysis: The limited processing capability and communication speed of the MG90S servo contribute to the response delay. The servo needs to receive the control signal, process it internally, and then drive the motor to rotate to the specified position. This series of processes takes a certain amount of time, and the low-cost and simple design of the MG90S servo make it difficult to achieve a fast response.

5.2 Improvement Solutions

Hardware Upgrade

- **Servo:** Replace it with the MG996R servo (torque 2.5kg·cm, price about 25 yuan each), which can provide greater torque and support higher sorting speeds.
- **Sensor:** Add a light-blocking hood to effectively improve the TCS34725 sensor's anti-light interference capability and further enhance the accuracy of color recognition.

Software Optimization

- **Parallel Control:** Perform color detection simultaneously while the top servo resets to reduce the system's idle time and improve sorting efficiency.
- **Dynamic Threshold:** Automatically adjust the RGB range according to the lighting intensity (e.g., if $r > g \cdot 1.2$), enabling the system to better adapt to different lighting environments.

5.3 Scalability Analysis

Target: 300 candy beans in 3 minutes

- **Feasibility:** The current system framework needs to be reconstructed (e.g., adding a conveyor belt for feeding), but the following optimization measures within the budget are expected to achieve the target:
 - Replace with MG996R servos, increasing the cost by about 40 yuan to improve servo performance.
 - Further optimize the code logic to reduce the delay time to 50ms and improve the system's operating speed.

5.4 Reflection on Dual-Servo Design Compared with Other Groups

In class, I saw that other groups achieved a sorting speed of 300 candy beans in 3 minutes, which triggered my in-depth reflection on my own design. After analysis, I believe that although the dual-servo design balances performance and cost to a certain extent, it has certain limitations when pursuing higher speeds. Other groups may have adopted more efficient mechanical structures (such as conveyor belt feeding) and more powerful hardware configurations (such as servos with higher torque), thus achieving higher sorting speeds. However, I think that the dual-servo design has unique advantages in terms of low cost and easy implementation, suitable for scenarios with low speed requirements. Within a limited budget, the dual-servo design can meet basic design goals through reasonable collaborative work and optimization and also has certain potential in terms of scalability. In the future, system performance can be further improved through hardware upgrades and software optimizations to ultimately achieve higher sorting speeds.

In addition, the teacher mentioned to me at that time that other groups had anti-sticking designs for candy beans.

Regarding the anti-sticking design for candy beans, we can make improvements from the following aspects:

- **Optimize the Slope Structure:** Redesign the shape and material of the slope, such as using a smoother surface material or increasing the inclination angle of the slope, but ensuring that the increase in the angle does not affect the servo's response speed.
- **Add Auxiliary Devices:** Add some auxiliary devices on the slope, such as a micro-vibrator. When a candy bean is detected to be stuck, the vibrator is activated to help the candy bean slide down. Or design a "brush-like" device to gently push the candy beans during the sliding process, reducing the possibility of sticking.

6. Conclusion

Achievements Summary

- Successfully implemented a low-cost (293.2 yuan) and high-precision (97.4% accuracy) candy bean sorting system, proving the feasibility and effectiveness of the design.
- The dual-servo collaborative design achieved a good balance between performance and cost, providing a guarantee for the stable operation of the system.

Future Directions

1. Introduce Machine Learning Technology

1. **Application Scenarios:** Implement machine learning models, such as those based on TensorFlow Lite, to dynamically adjust color thresholds. This technology can be trained with a diverse dataset of candy bean colors under various lighting conditions, enabling the system to adapt to complex and changing environments. For instance, it can be used to distinguish between subtle color variations in different brands of candy beans or to handle color shifts due to aging or manufacturing differences.
2. **Expected Effects:** By leveraging machine learning, the system will achieve higher accuracy in color recognition, especially in challenging lighting scenarios. This will reduce the need for manual threshold calibration and enhance the system's overall robustness and adaptability.

2. Expand into a Multi-Channel Sorting System

1. Implementation Scheme:

1. **Increase Servos:** Add additional servos to control multiple sorting channels. For example, four servos could be used to manage four distinct sorting paths, each dedicated to a specific color category.

2. **Conveyor Belt Integration:** Incorporate a conveyor belt system for continuous and automated feeding of candy beans into the sorting mechanism. This will eliminate the need for manual placement and significantly increase the sorting throughput.
 3. **Enhanced Control Logic:** Develop advanced control algorithms to manage the coordination between the conveyor belt, the color sensor, and the multiple servos. This will ensure smooth and efficient sorting operations, even at higher speeds.
2. **Expected Effects:** The multi-channel sorting system will dramatically increase the sorting speed, potentially reaching the target of 300 candy beans in 3 minutes. It will also enhance the system's flexibility, allowing it to handle a wider variety of candy bean colors and types with ease.

3. Improve Anti-Sticking Design

1. **Optimization Measures:**
 1. **Slope Redesign:** Use a smoother and more slippery surface material for the slope, such as a polished metal or a Teflon-coated surface, to reduce friction and prevent candy beans from sticking.
 2. **Auxiliary Devices:** Install a micro-vibrator or a small fan on the slope to create gentle vibrations or airflow that can help dislodge any stuck candy beans. Alternatively, design a "brush" or "wiper" mechanism to gently push the candy beans along the slope when necessary.
2. **Expected Effects:** These improvements will significantly reduce the occurrence of candy beans getting stuck on the slope, ensuring continuous and efficient sorting operations. They will also enhance the system's reliability and reduce the need for manual intervention.

4. Add Energy-Saving Design

1. **Implementation:**
 1. **Sorting Completion Detection:** Incorporate a sensor to detect when the last candy bean has been sorted. Once detected, the system can automatically shut down or enter a low-power mode to conserve energy.
 2. **Power Management Module:** Implement a power management module that can monitor and optimize the system's power consumption. This could include features like automatic sleep modes during idle periods and efficient power distribution to different components.

2. **Expected Effects:** The energy-saving design will reduce the system's operational costs and environmental impact by minimizing power usage. It will also extend the lifespan of the system's components by reducing wear and tear caused by continuous operation.

7. Appendix

Test Data Table

Time (Minutes)	Number of Sorted Candy Beans	Accuracy	Remarks
3	76	97.4%	2 errors due to bottom servo delay

Key Problem Analysis

- **Why Choose MG90S Instead of 360° Servos?**
 - The 180° servo (MG90S) has sufficient precision to meet the sorting requirements and is low-cost. A 360° servo requires an additional encoder for position control, which not only increases the cost but also violates the principle of low-cost design.
- **What Is the Root Cause of Sorting Errors?**
 - The servo actions were not synchronized, and the code lacked the necessary delay() waiting time, resulting in the bottom servo not responding in time, rather than a problem with the sensor itself.
- **How to Increase the Speed to 300 Candy Beans in 3 Minutes?**
 - It is necessary to upgrade the servos (such as to MG996R) to provide greater torque and optimize the code logic to reduce the delay to 50ms. Within the budget, the above improvements (MG996R + light-blocking hood $\approx$ 60 yuan increment) are feasible.

References

- [1] Yuan Qingsong. Design of a Sorting System for Production Line Materials Based on a Vision Manipulator [D]. Guangxi Normal University, 2024. DOI:10.27036/d.cnki.ggxsu.2024.001294.
- [2] Xu Xiaoyan, Cui Xueying. Design of a Color Recognition and Sorting System Based on a Single-Chip Microcomputer [J]. Science and Technology & Innovation, 2021, (21): 109-110+113. DOI:10.15913/j.cnki.kjyex.2021.21.047.
- [3] Wang Yangyang. Design of a Sorting System for Waste Plastic Bottles Based on Vision and Industrial Robots [D]. Yancheng Institute of Technology, 2023. DOI:10.44381/d.cnki.gycit.2023.000209.
- [4] Wu Jiahong, Hu Xuefei, Guo Junhao, et al. Intelligent Color Sorter Based on Jetson Nano Platform and Manipulator [J]. Electronic Production, 2023, 31(02): 34-36+25. DOI:10.16589/j.cnki.cn11-3571/tn.2023.02.011.